COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S1
DATE PERFORMED:	07/09/2026	**DATE SUBMITTED:**	
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

5. Procedure

ILO A: Create Code That Follows the Basic C++ Code Structure

For this activity, you have to demonstrate the use of a function prototype. The section on class declaration and method definition will be used for the function prototype and the function will be defined in the follow method definition section after the main function.
A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

Create a function that will take two numbers and display the sum.

Header File Declaration Section:

```
#include <iostream>
```

Global Declaration Section:

```
int SumofTwoNum(int num1, int num2);
```

Class Declaration and Method Definition Section:

```
int SumofTwoNum(int fNum, int lNum) {
    int total = fNum + lNum;
    return total;
}
```

Main Function:

```
int main() {
    std::cout << SumofTwoNum(6, 14) << std::endl;

    return 0;
}
```

Create a function that will return whether variable A is greater than variable B.

Header File Declaration Section:

```
#include <iostream>
```

Class Declaration and Method Definition Section:

```
bool GreaterorLess(int a, int b){
  return a > b;
}
```

Main Function:

```
int main(){
  int a, b;

  std::cout << "Number A: ";
  std::cin >> a;

  std::cout << "Number B: ";
  std::cin >> b;

  if (GreaterorLess(a, b))
    std::cout << "A is Greater than B";
  else if (a == b)
    std::cout << "A is Equal to B";
  else
    std::cout << "A is Less than B";

  return 0;
}
```

Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Header File Declaration Section:

```
#include <iostream>
```

Global Declaration Section:

```
bool showLogicalOperations(bool a, bool b);
```

Main Function:

```
int main() {

    showLogicalOperations(true, false);

    return 0;
}
```

Method Definition:

```
12
13  bool showLogicalOperations(bool a, bool b) {
14
15      std::cout << "Type 1 if true and 0 if false" << std::endl;
16      std::cin >> a;
17      std::cin >> b;
18
19
20      std::cout << "A AND B: " << (a && b) << std::endl;
21      std::cout << "A OR B:  " << (a || b) << std::endl;
22      std::cout << "NOT A:   " << (!a) << std::endl;
23      std::cout << "NOT B:   " << (!b) << std::endl;
24      std::cout << "A XOR B: " << (a ^ b) << std::endl;
25      std::cout << "A NAND B: " << !(a && b) << std::endl;
26      std::cout << "A NOR B: " << !(a || b) << std::endl;
27      std::cout << "A XNOR B: " << !(a ^ b) << std::endl;
28
29
30      return true;
31  }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\VSCode\Projects> cd "d:\VSCode\Projects\EGAR_ILO-A_num3\" ; if ($?) { g++
EGAR_ILO-A_num3 } ; if ($?) { .\EGAR_ILO-A_num3 }
Type 1 if true and 0 if false
1
0
A AND B: 0
A OR B:  1
NOT A:   0
NOT B:   1
A XOR B: 1
A NAND B: 1
A NOR B: 0
A XNOR B: 0
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

EGAR_ILO-B > EGAR_ILO-B.cpp > ...

```cpp
#include <iostream>

class Triangle {
    private:
        double totalAngle, angleA, angleB, angleC;

    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        bool ValidateTriangle() const;
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = angleA + angleB + angleC;
}
```

```cpp
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = angleA + angleB + angleC;
}

bool Triangle::ValidateTriangle() const {
    return totalAngle == 180;
}
```

```cpp
int main() {
    Triangle set1(40, 30, 110);
    if (set1.ValidateTriangle()) {
        std::cout << "The triangle is valid." << std::endl;
    } else {
        std::cout << "The triangle is not valid." << std::endl;
    }
    return 0;
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
_ILO-B } ; if ($?) { .\EGAR_ILO-B }
The triangle is valid.
PS D:\VSCode\Projects\EGAR_ILO-B> cd "d:\VSCode\Projects\EGAR_ILO-B\" ; if ($?) { g++
_ILO-B } ; if ($?) { .\EGAR_ILO-B }
The triangle is valid.
PS D:\VSCode\Projects\EGAR_ILO-B> cd "d:\VSCode\Projects\EGAR_ILO-B\" ; if ($?) { g++
_ILO-B } ; if ($?) { .\EGAR_ILO-B }
The triangle is not valid.
PS D:\VSCode\Projects\EGAR_ILO-B> cd "d:\VSCode\Projects\EGAR_ILO-B\" ; if ($?) { g++
_ILO-B } ; if ($?) { .\EGAR_ILO-B }
The triangle is not valid.
PS D:\VSCode\Projects\EGAR_ILO-B> cd "d:\VSCode\Projects\EGAR_ILO-B\" ; if ($?) { g++
_ILO-B } ; if ($?) { .\EGAR_ILO-B }
The triangle is valid.
PS D:\VSCode\Projects\EGAR_ILO-B>
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.

EGAR_SUPPLE-1.cpp > main()

```cpp
#include <iostream>

// swap the two numbers in different variables.

int num1, num2, temp;

int main() {

    std::cout << "Input first number: ";
    std::cin >> num1;
    std::cout << "Input second number: ";
    std::cin >> num2;

    temp = num1;
    num1 = num2;
    num2 = temp;

    std::cout << "After swapping, first number: " << num1 << std::endl;
    std::cout << "After swapping, second number: " << num2 << std::endl;

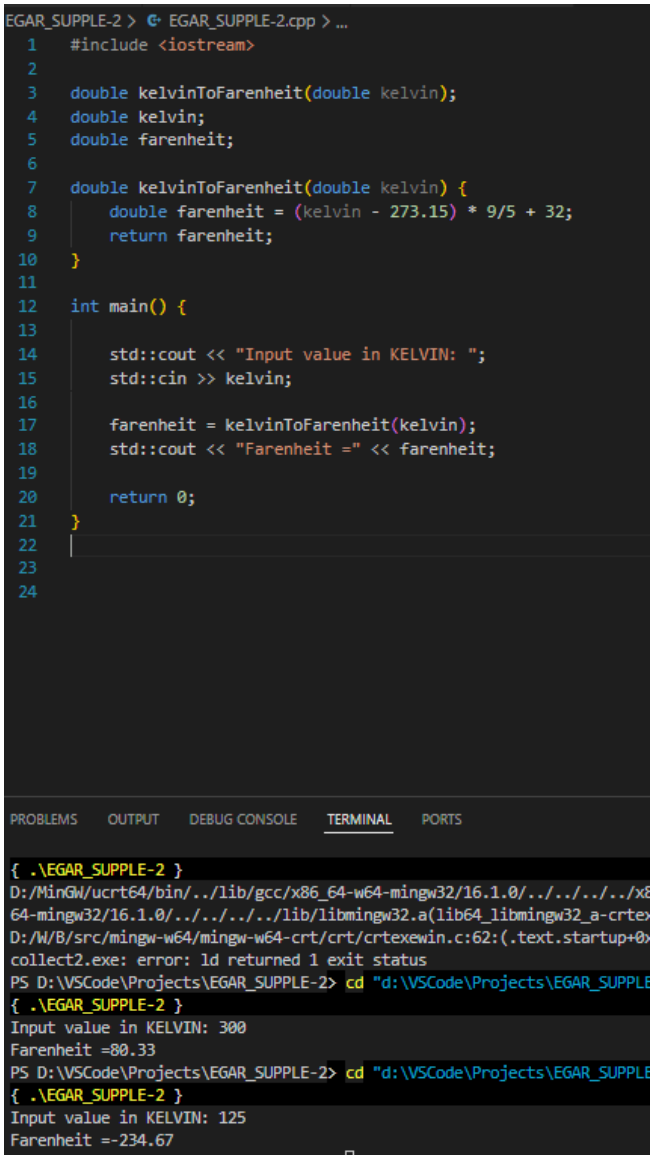
    return 0;

}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\VSCode\Projects> cd "d:\VSCode\Projects\" ; if ($?) { g++ EGAR_SUPPLE-1.cpp -o EGAR_SUPPLE-1 } ; if ($?) { .\EGAR_SUPPLE-
Input first number: 3
Input second number: 1
After swapping, first number: 1
After swapping, second number: 3
PS D:\VSCode\Projects>
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.



```
EGAR_SUPPLE-2 > EGAR_SUPPLE-2.cpp > ...
  1  #include <iostream>
  2
  3  double kelvinToFarenheit(double kelvin);
  4  double kelvin;
  5  double farenheit;
  6
  7  double kelvinToFarenheit(double kelvin) {
  8      double farenheit = (kelvin - 273.15) * 9/5 + 32;
  9      return farenheit;
 10  }
 11
 12  int main() {
 13
 14      std::cout << "Input value in KELVIN: ";
 15      std::cin >> kelvin;
 16
 17      farenheit = kelvinToFarenheit(kelvin);
 18      std::cout << "Farenheit =" << farenheit;
 19
 20      return 0;
 21  }
 22
 23
 24
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

{ .\EGAR_SUPPLE-2 }
D:/MinGW/ucrt64/bin/../lib/gcc/x86_64-w64-mingw32/16.1.0/../../../../x8
64-mingw32/16.1.0/../../../../lib/libmingw32.a(lib64_libmingw32_a-crtex
D:/W/B/src/mingw-w64/mingw-w64-crt/crt/crtexewin.c:62:(.text.startup+0x
collect2.exe: error: ld returned 1 exit status
PS D:\VSCode\Projects\EGAR_SUPPLE-2> cd "d:\VSCode\Projects\EGAR_SUPPLE
{ .\EGAR_SUPPLE-2 }
Input value in KELVIN: 300
Farenheit =80.33
PS D:\VSCode\Projects\EGAR_SUPPLE-2> cd "d:\VSCode\Projects\EGAR_SUPPLE
{ .\EGAR_SUPPLE-2 }
Input value in KELVIN: 125
Farenheit =-234.67
```

3. Create a C++ program that has a function that will calculate the distance between two points.

EGAR_SUPPLE-3 > EGAR_SUPPLE-3.cpp > ...

```cpp
#include <iostream>
#include <cmath>

double a1, b1, a2, b2;
double distance;

int main() {

    std::cout << "Input coordinates x1: ";
    std::cin >> a1;
    std::cout << "Input coordinates y1: ";
    std::cin >> b1;
    std::cout << "Input coordinates x2: ";
    std::cin >> a2;
    std::cout << "Input coordinates y2: ";
    std::cin >> b2;

    distance = sqrt(pow(a2 - a1, 2) + pow(b2 - b1, 2));
    std::cout << "Distance between the points: " << distance << std::endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
      |                                    ~~ ^ ~~
      |                                    |    |
      |                                    |    double(double)
      |                                    double
PS D:\VSCode\Projects\EGAR_SUPPLE-3>


                                  > cd "d:\VSCode\Projects\EGAR_SUPPLE-3\" ; if ($?) { g++ EGAR_SUPPL
 -o EGAR_SUPPLE-3 } ; if ($?) { .\EGAR_SUPPLE-3 }
Input coordinates x1: 32.1
Input coordinates y1: 125.0
Input coordinates x2: 125.163
Input coordinates y2: 3269.0
Distance between the points: 3145.38
PS D:\VSCode\Projects\EGAR_SUPPLE-3>
```

4. Modify the code given in ILO B and add the following functions:

EGAR_SUPPLE-4 > EGAR_SUPLLE-4.cpp > determineTheType() const

```cpp
#include <iostream>
#include <cmath>
#include <string>

class Triangle {
    private:
        double totalAngle, angleA, angleB, angleC, sideA, sideB, sideC;

    public:
        Triangle(double A, double B, double C, double sdA = 0, double sdB = 0, double sdC = 0);
        void setAngles(double A, double B, double C);
        void setsides(double sA, double sB, double sC);
        bool ValidateTriangle() const;

        // question 4
        double computeTheArea() const;
        double computeThePerimeter() const;
        std::string determineTheType() const;
};

Triangle::Triangle(double A, double B, double C, double sdA, double sdB, double sdC) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = angleA + angleB + angleC;

    sideA = sdA;
    sideB = sdB;
    sideC = sdC;
}
```

```cpp
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = angleA + angleB + angleC;
}


void Triangle::setsides(double sA, double sB, double sC) {
    sideA = sA;
    sideB = sB;
    sideC = sC;
}

bool Triangle::ValidateTriangle() const {
    return totalAngle == 180;
}
```

a. A function to compute for the area of a triangle

```
// question 4.a
double Triangle::computeTheArea() const {
    double s = computeThePerimeter() / 2;
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}
```

b. A function to compute for the perimeter of a triangle

```
// question 4.b
double Triangle::computeThePerimeter() const {
    return sideA + sideB + sideC;
}
```

c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```
// question 4.c
std::string Triangle::determineTheType() const {
    if (angleA < 90 && angleB < 90 && angleC < 90) {
        return "acute-angled";
    }
    else if (angleA > 90 || angleB > 90 || angleC > 90) {
        return "obtuse-angled";
    }
    else {
        return "others";
    }
}
```

```cpp
int main() {

    Triangle set1(40, 30, 110, 7.0, 5.5, 10.2);

    if (set1.ValidateTriangle()) {
        std::cout << "The triangle is valid." << std::endl;


        std::cout << "Area: " << set1.computeTheArea() << std::endl;
        std::cout << "Perimeter: " << set1.computeThePerimeter() << std::endl;
        std::cout << "Type: " << set1.determineTheType() << std::endl;

    } else {
        std::cout << "The triangle is not valid." << std::endl;
    }
    return 0;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\VSCode\Projects> cd "d:\VSCode\Projects\EGAR_SUPPLE-4\" ; if ($?) { g++ EGAR_SUPLLE-4.cpp -o
The triangle is valid.
Area: 18.2251
Perimeter: 22.7
Type: obtuse-angled
PS D:\VSCode\Projects\EGAR_SUPPLE-4>
```

Summary for procedures:
The procedure helped me to recall my past understanding of c++, although I had trouble in recalling most of them but by the help of the instructions, it helped me understand what I need to do.

Summary for Supplementary:
The supplementary questions helped me to use what I recall during the procedures, however I still had a hard time coding the number 1 and 4 because I was confused on what to do first. So I had to

Conclusion:
Overall this HOA has helped me regain my knowledge of c++ and the process of understanding the problem for me to start the code.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Fixing VS Code, Step-by-step process for the code**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **My program was having an error, although it was correct, and I tried running it in an online compiler and it worked. So, I asked Gemini how to fix it, and it turns out that I just have to restart VS Code for it to work. Lastly I asked Gemini to help me understand what I should do first and what I should be thinking to better understand how to properly start the code. But I told gemini to not give me the actual code so that I can practice myself.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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